System Protection Thermal Analysis

Project: Automatic Power Factor Correction Meter

Department of Electrical Engineering, NUST SEECS

Technical Calculation Sheet

Project: Power Factor Correction Meter

Department of Electrical Engineering, NUST SEECS

1 Section 1: Bleeder Resistor Heat Dissipation

To ensure user safety, bleeder resistors are connected in parallel to the capacitor banks to discharge stored energy upon system de-energization.

1.1 Component Specifications

The following parameters are used for the safety analysis of the Capacitor Bank discharge circuit:

- **Capacitance (C):** $2.2\,\mu$F (Rated for 400 V, 50 Hz)
- **Bleeder Resistors (R_1, R_2):** $27\,\mathrm{k}\Omega$ each (Rated for 1 W)
- **Configuration:** R_1 and R_2 are in series, connected in parallel across the capacitor.
- **Input Voltage ($V_{\mathbf{rms}}$):** 230 V AC (Standard operating voltage)

1.2 Resistance and Power Dissipation Calculations

1.3 Total Bleeder Resistance

Since the two resistors are connected in series, the total resistance R_{total} is:

$$R_{\text{total}} = R_1 + R_2 = 27\,\mathrm{k}\Omega + 27\,\mathrm{k}\Omega = 54\,\mathrm{k}\Omega \tag{1}$$

1.4 Steady-State Power Dissipation (Heat)

The total power dissipated by the resistors while the circuit is energized is calculated using the RMS voltage:

$$P_{\text{total}} = \frac{V_{\text{rms}}^2}{R_{\text{total}}} = \frac{230^2}{54,000} \approx 0.9796\,\mathrm{W} \tag{2}$$

1.5 Stress Analysis per Resistor

Because the resistors are identical and in series, the power is shared equally:

$$P_{\text{resistor}} = \frac{P_{\text{total}}}{2} = \frac{0.9796}{2} \approx 0.4898\,\text{W} \tag{3}$$

Safety Conclusion: Each resistor is dissipating $\approx 0.49\,\text{W}$. Given their $1\,\text{W}$ rating, the resistors are operating at **49% of their maximum capacity**, providing a safe thermal margin for continuous operation.

1.6 Discharge Time Analysis

For safety during maintenance, the discharge time constant (τ) determines how quickly the stored energy is neutralized after disconnection.

$$\tau = R_{\text{total}} \times C = 54,000\,\Omega \times 2.2 \times 10^{-6}\,\text{F} = 0.1188\,\text{s} \tag{4}$$

The time required to reach a safe voltage threshold ($< 50\,\text{V}$) is:

$$t_{\text{safe}} = -\tau \cdot \ln\left(\frac{V_{\text{safe}}}{V_{\text{peak}}}\right) = -0.1188 \cdot \ln\left(\frac{50}{230\sqrt{2}}\right) \approx 0.222\,\text{s} \tag{5}$$

1.7 Summary Table

Parameter	**Calculated Value**
Total Series Resistance	$54\,\text{k}\Omega$
RMS Power Loss (Heat)	$0.98\,\text{W}$
Power per Resistor	$0.49\,\text{W}$
Thermal Utilization	49% (Safe)
Discharge Time to Safe Level	$\approx 0.22\,\text{s}$

2 Section 2: Inrush Current Limitation (Sand Block Analysis)

High-wattage "Sand Block" resistors are placed in series with the capacitors to limit the instantaneous current surge during relay switching.

2.1 Case A: 4.7 Ω (10W) with 1.1 μF Capacitor

The peak inrush current occurs if switching happens at V_{peak} (325.3 V):

$$I_{peak} = \frac{325.3\,\mathrm{V}}{4.7\,\Omega} \approx \mathbf{69.21}\,\mathrm{A} \tag{6}$$

The charging energy absorbed by the resistor per switch is **0.058 J**.

2.2 Case B: 12 Ω (10W) with 2.2 μF Capacitor

$$I_{peak} = \frac{325.3\,\mathrm{V}}{12\,\Omega} \approx \mathbf{27.11}\,\mathrm{A} \tag{7}$$

The charging energy absorbed by the resistor per switch is **0.116 J**.

3 Summary Data Table

Component	Load Type	Parameter	Value
Bleeder Pair	2.2 μF	Operating Temp Rise	Moderate ($\approx$ 50%)
Sand Block A	1.1 μF	Max Surge Current	69.2 A
Sand Block B	2.2 μF	Max Surge Current	27.1 A

Engineering Conclusion:
The use of 10W sand block resistors provides a high thermal mass that easily absorbs the microsecond current pulses during switching. The 12 Ω resistor is particularly effective at extending relay life by keeping the surge current under 30A.

Calculated by: Ammar, Ali, Rana Hamza, Uzair, Waqas

4 Section 4: Analytical Determination of Correction Capacitance

To achieve the target phase angle of 15° while maintaining the measured system power of 60.79 W, the required capacitance is derived below.

4.1 Reactive Power Requirement

The uncorrected system displays a calculated natural lag of $\phi_1 \approx 47.7°$. To reduce this to the target $\phi_2 = 15°$, the necessary capacitive reactive power (Q_c) is:

$$Q_c = P(\tan 47.7° - \tan 15°) = 60.79(1.0996 - 0.2679) \approx 50.56\,\text{VAR} \tag{8}$$

4.2 Capacitance Required

Using the fundamental relation $Q_c = V^2\omega C$ with $V = 220.78\,\text{V}$ and $f = 50\,\text{Hz}$:

$$C = \frac{Q_c}{V^2 \cdot 2\pi f} = \frac{50.56}{(220.78)^2 \cdot 314.16} \tag{9}$$

$$\mathbf{C} \approx \mathbf{3.3}\,\mu\text{F} \tag{10}$$

4.3 Conclusion

The calculated value aligns precisely with the experimental $3.3\,\mu$F capacitor bank utilized in the hardware implementation. This confirms that the selected capacitance provides the exact reactive compensation needed to reach a power factor of 0.966 (lag) for the current laboratory load.

5 Zero-Crossing Detection and Signal Conditioning

The APFC system utilizes a comparator-based Zero-Crossing Detector (ZCD) to generate a timing reference. The following derivation defines the switching threshold.

5.1 Static Reference Derivation

The non-inverting input V_+ is clamped by a forward-biased diode supplied by a 12 V DC source. The reference voltage is:

$$V_+ = V_{diode} \approx 0.5\,\mathrm{V} \tag{11}$$

5.2 Dynamic Signal Transfer Function

The pulsating DC input V_{sig} is processed through a voltage divider to the inverting input V_-:

$$V_- = V_{sig}(t) \cdot \frac{1\,\mathrm{k\Omega}}{1\,\mathrm{k\Omega} + 3.3\,\mathrm{k\Omega}} = 0.232 \cdot V_{sig}(t) \tag{12}$$

5.3 Switching Threshold and Pulse Width

The comparator output V_{out} transitions when $V_+ = V_-$. Solving for the input signal voltage V_{sig} at the transition point:

$$0.5 = 0.232 \cdot V_{sig} \implies V_{sig} \approx 2.1\,\mathrm{V} \tag{13}$$

As the rectified AC wave drops below 3.02 V, the output pulses High. This 100 Hz pulse train provides the base frequency for the microcontroller's phase-shifting algorithm.